

Parishram 2027
Biology
Principles Of Inheritance and Variation

DPP: 3

Q1 Which of the following statements are correct about incomplete dominance?

1. The F_1 phenotype is intermediate between the two parental phenotypes
2. F_2 phenotypic and genotypic ratios are the same (1 : 2 : 1)
3. Incomplete dominance occurs due to the blending of genes
4. It violates Mendel's principle of dominance

- (A) 1 and 2 only
(B) 1, 2, and 4 only
(C) 2 and 3 only
(D) All are correct

Q2 Assertion (A): In codominance, both alleles express themselves independently in heterozygous condition.

Reason (R): This results in an intermediate phenotype like in the case of pink flowers in *Mirabilis Jalapa*

- (A) A and R are true; R is the correct explanation of A
(B) A and R are true; R is not the correct explanation of A
(C) A is true, R is false
(D) A is false, R is true

Q3 A cross between a red-flowered (RR) and white-flowered (rr) *Mirabilis Jalapa* plant results in all pink-flowered plants in F_1 . What will be the phenotypic ratio in the F_2 generation?

- (A) 3 red: 1 white
(B) 1 red: 2 pink: 1 white

- (C) 2 red: 1 pink: 1 white
(D) All pink

Q4 Which of the following features is/are true regarding multiple alleles?

1. More than two alleles exist in a population.
2. An individual can carry all multiple alleles.
3. Example includes coat colour in rabbits.
4. ABO blood groups are due to multiple alleles

- (A) 1, 2, and 3
(B) 1, 3, and 4
(C) 2 and 4 only
(D) All are correct

Q5 Match the Following

Column A

A) Sickle Cell Anaemia

B) Phenylketonuria

C) Marfan Syndrome

Column B

1) Affects shape of RBC, causes anemia

2) Mental retardation, skin pigmentation

3) Affects eyes, heart, and skeleton

(A) A-1, B-2, C-3

(C) A-2, B-3, C-1

(B) A-3, B-1, C-2

(D) A-1, B-3, C-2

Q6 In a dihybrid cross involving seed shape (round/dominant, wrinkled /recessive) and seed color (yellow/dominant, green/recessive), what phenotypic ratio is expected in F_2 ?

- (A) 3 : 1
(B) 9 : 3 : 3 : 1
(C) 1 : 2 : 1
(D) 1 : 1



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Q7 In ABO blood grouping, if a child has blood group AB, which of the following parental combinations is not possible?

- (A) Father – A, Mother – B
- (B) Father – AB, Mother – O
- (C) Father – AB, Mother – AB
- (D) Father – B, Mother – A

Q8 In incomplete dominance, the heterozygous genotype expresses a phenotype that is _____ to both homozygous conditions

- (A) Identical
- (B) Superior
- (C) Intermediate
- (D) Recessive

Q9 Identify the incorrect statement

- (A) Co-dominance results in expression of both alleles independently.
- (B) In pleiotropy, a gene can affect multiple traits.
- (C) Incomplete dominance leads to blending of genes
- (D) Multiple alleles can be found in a population but not more than two in an individual

Q10 A tall pea plant with axial flowers (TtAa) is test-crossed with a dwarf pea plant with terminal flowers (ttaa). What percentage of offspring are expected to be tall with terminal flowers?

- (A) 25%
- (B) 50%
- (C) 12.5%
- (D) 75%



Answer Key

Q1 B
Q2 C
Q3 B
Q4 B
Q5 A

Q6 B
Q7 B
Q8 C
Q9 C
Q10 A



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Hints & Solutions

Note: scan the QR code to watch video solution

Q1 Text Solution:

In incomplete dominance, alleles do not blend but rather both influence the phenotype. The genes retain their identity

Video Solution:



Q2 Text Solution:

The Reason describes incomplete dominance, not codominance. In codominance, like in ABO blood groups, both alleles are fully expressed.

Video Solution:



Q3 Text Solution:

In incomplete dominance, F_2 shows a 1 : 2 : 1 phenotypic as well as genotypic ratio.

Video Solution:



Q4 Text Solution:

Statement 2 is incorrect—only two alleles can exist in an individual, even if the population has multiple alleles

Video Solution:



Q5 Text Solution:

These are all examples of pleiotropy, where one gene affects multiple traits.

Video Solution:



Q6 Text Solution:

This is a classic Mendelian dihybrid cross, showing independent assortment.

Video Solution:



Q7 Text Solution:

O group individual has genotype ii and cannot pass I or I allele required to form AB.

Video Solution:



Q8 Text Solution:

Pink flower is intermediate between red and white in *Mirabilis jalapa*.



[Android App](#)

| [iOS App](#)

| [PW Website](#)

Video Solution:



Q9 Text Solution:

In incomplete dominance, phenotypes blend, not the genes. Alleles remain unaltered.

Video Solution:



Q10 Text Solution:

Explanation: Test cross $TtAa \times ttaa$

Gametes: $TA, Ta, tA, ta \times ta$

Tall (T-) + terminal (aa): only $Ta \times ta$ 25%

Video Solution:



[Android App](#)



[iOS App](#)



[PW Website](#)